Does the job market influence voting brackets ?
Correlation between unemployment metrics and themes in political manifestos.

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Abstract

This paper evaluates correlations between unemployment themes present in political manifestos of French national elections and the real state of unemployment at the time and place where the candidates were running for office. Overall, correlations are dim, except for the far left parties, who exhibit both more unemployment related manifestos, and higher correlation with unemployment realities than all the other political parties.

1 Introduction and problem statement

In 1978, the French far right party *Front National* wrote "1 million de chômeurs, c'est 1 million d'immigrés de trop" (in english : "1 million unemployed is 1 million immigrants too many") on their political leaflets in anticipation for the legislative elections of that year. This rather simplified equation serves as an example of political parties surfing on economical trends to create political discourses. In the case of the *Front National*, to serve their anti-immigration objectives [5]. In France (and in many other different countries), the oil-related economic crises of 1973 and 1975 provoked a sudden increase in unemployment that tripled the number of people unemployed between 1967 and 1978. In 1988, the unemployment rate was of 8,4% and increased very fast between 1990 and 1993 leading it to be of 9,6% in 1993 (source : Insee). Due to this sudden increase, unemployment was becoming a major political topic for voters and so for politicians. This is what has driven us to conduct this study based on political manifestos and leaflets of the legislative elections of 1993 in France. The question we intend to answer is whether unemployment ("chômage") is a key topic in this election, and if it the use of this topic is consistent with economical realities. Importantly, this analysis is done at a local level seeing as unemployment differs between French departments and that legislative elections mobilize local subjects. With this local analysis, we wonder if mention of unemployment in political leaflets is correlated to effective unemployment rates and whether this correlation differs through different political parties.

2 State of the art : NLP for politics

This approach is important to verify some political science theories on the responsiveness of voters to economic context. In fact political science theory adresses this question, in particular through the lens of *economic voting* theorized by M.S. Lewi-Beck and M. Stegmaier [7]. Economic issues matter to voters and voters sanction or celebrate their local politicians in regard of economic evolution. This topic of political science can be analyzed with quantitative methods, which is the case of an increasing number of topics such as suggested by [3] which describes the advantages that NLP can have for the analysis of political discourse. Analysis of political discourse is always confronted to the issue of prepared political discourse, not always very significant ("langue de bois") [6].

*See the github for the project at this adress : link.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

To study the link between political discourse on unemployment and effective unemployment, we rely on the 1993 legislative elections corpus from Archelec, a digital archive of French electoral leaflets compiled by Sciences Po [2]. The corpus contains 2120 leaflets for 1993, covering all French departments. We begin by using LDA [1] as a classic NLP method to detect salient topics in the 1988 and 1993 manifestos. Due to some limits of the LDA method and in order to account for the nuances of political rhetoric, we implement the BERTopic method [4]. The topic intensities obtained for the topic of employment are then matched with department-level unemployment rates from Insee to assess the relationship between local economic conditions and electoral discourse.

3 Methodology

3.1 Data used for the project

3.1.1 Retrieving and preprocessing

DISCLAIMER

A small disclaimer : whilst breezing through the website, I didn't realize that full and clean transcriptions were available on a git repository. I consequently designed a pipeline for transcription retrieval using requests at the `InternetArchives.org` website, but was faced with partial results. It was however an interesting opportunity to dig into text cleaning techniques.

The analysis focuses on the 1993 legislative elections in France. Upon data OCR retrieval on the `InternetArchives.org` website, the pre-cleaning sample consists of $N = 5837$ entries. However, part of the text is unusable, because of OCR failures, and consists of garbage text.

Example of OCR Garbage

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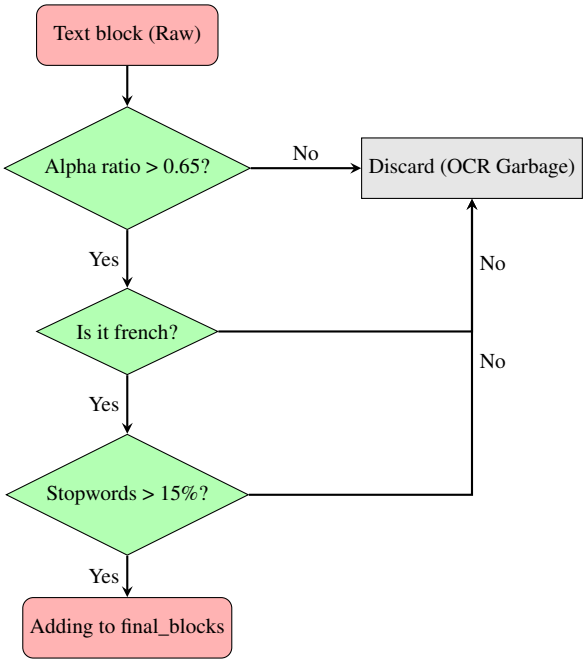
To account for this problem, a three-step text cleaning pipeline was implemented, described in Figure 1.

Following this preprocessing pipeline, the final amount of exploitable text is $N = 2120$, or 36.3% of the original corpus.

3.1.2 Preliminary data analysis

As displayed on Figure 2, a preliminary analysis of the structure of candidatures was conducted. The first observation is that, for age and profession, a lot of data is missing (respectively $NaN = 1326$ and $NaN = 1098$). However, we can see that the majority of reported values draw a certain portrait of the typical candidate for the legislative election : a male, in the prime of life, belonging to higher socio-economical background (with either strong cultural or economical capitals). Paris and other Île-de-France departments concentrate the most candidates, followed by the north of France, the East, and the South-West of France.

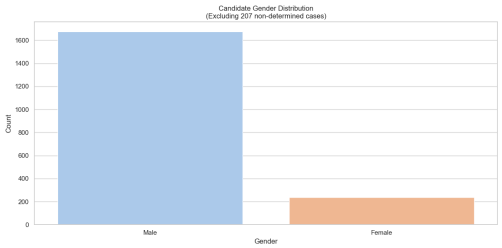


Data Cleaning Pipeline

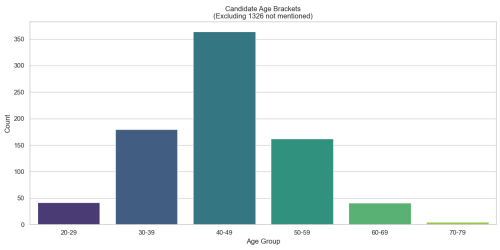
OCR text is divided into blocks of text, which are then all analyzed with the following pipeline :

- **Alpha Ratio:** Looking at the ratio of letters over other characters, this step filters out blocks dominated by symbols or numbers.
- **Language Check:** A language detection model is ran over the text block (using the `fasttext` library), and filters out blocks whose french probability are under 60%.
- **Stopwords Density:** Finally, blocks that display a low level of common french stopwords usefull for phrase building are filtered out, using the `spacy` library.

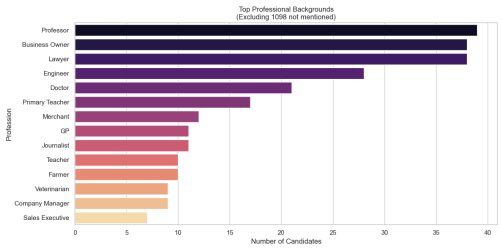
Figure 1: Preprocessing workflow for clean text detection.



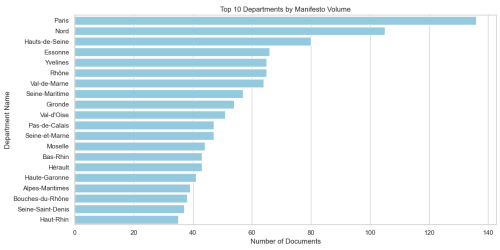
(a) Candidate Gender Distribution



(b) Candidate Age Brackets



(c) Top Professional Backgrounds



(d) Top 20 Departments by Volume

Figure 2: Descriptive statistics of the candidate dataset (2002–2022).

3.2 Models for topic modeling

To conduct the topic analysis, two approaches were chosen, each working with different assumptions and statistical mechanisms.

3.2.1 Latent Dirichlet Allocation

Latent Dirichlet Allocation, is a probabilistic topic modeling technique whose core idea is that documents are constituted as a mix of topics, where each topic is defined as a mix of words. LDA computes the probability of a word belonging to a topic, and a document containing a topic, relying on word co-occurence and frequency.

3.2.2 BERTopic

BERTopic on the other hand relies on text vector embeddings to grasp the semantic meaning of text data rather than a purely syntactical approach. The process begins with embedding documents using transformer models like BERT, than reducing the dimension of resulting vectors prior to clustering, than identifying dense group of documents based on the reduced vector.

4 Results

4.1 Topic modeling

4.1.1 LDA results

The LDA model trained on the whole corpus of texts identifies the following 10 themes (table 1) :

ID	Top Keywords
0	français, immigration, retraite, maintenir, impôt, naturel
1	mars, falloir, pouvoir, homme, pays, circonscription
2	communiste, vote, pouvoir, entendre, peuple, parti
3	droite, gauche, écologiste, force, voter, vouloir
4	tour, dimanche, électeur, candidat, majorité, circonscription
5	travailleur, faire, entreprise, emploi, payer, patron
6	français, renaissance, ingénieur, devoir, prochain, défense
7	environnement, vie, nouveau, emploi, développement, solidarité
8	français, voix, élu, traité, entendre, taxe
9	écologie, carte, argent, renouvelable, incapable, prétendre

Table 1: LDA topic modeling results (Top 6 words per topic).

The topics cover a broad thematic range: general political rhetoric (Topic 0), electoral process language (Topics 1, 4), left-wing and communist discourse (Topics 2, 3), social and environmental policy (Topic 7), national identity and far-right discourse (Topic 8), and ecology (Topic 9). **Our topic of interest, treating of employment, topic 5, is the dominant topic in only 83 out of 2,120 manifestos, representing 3.9% of the corpus**. When it is the main topic, it accounts for an average probability of 0.72, indicating a strong and coherent signal within those documents. This very low rate of manifestos treating essentially of (un)employment is a strong limitation since the vast majority of manifestos in our corpus do not primarily discuss unemployment, which limits the statistical power of the correlation analyses. It does not mean that the topic is absent seeing as LDA assigns a non-zero probability to all topics for every document, but rather that unemployment is rarely the central theme.

4.1.2 BERTopic results

With the BERTopic model, texts are not assigned to every topic but rather to one single cluster out of the 19 detected by the model (including the noise cluster -1), see table 2. Only 26 documents (1.2% of the corpus) are assigned to the noise cluster, suggesting that the model is satisfying overall. However a very striking element is that topic 0 is excessively dominant by capturing 85% of

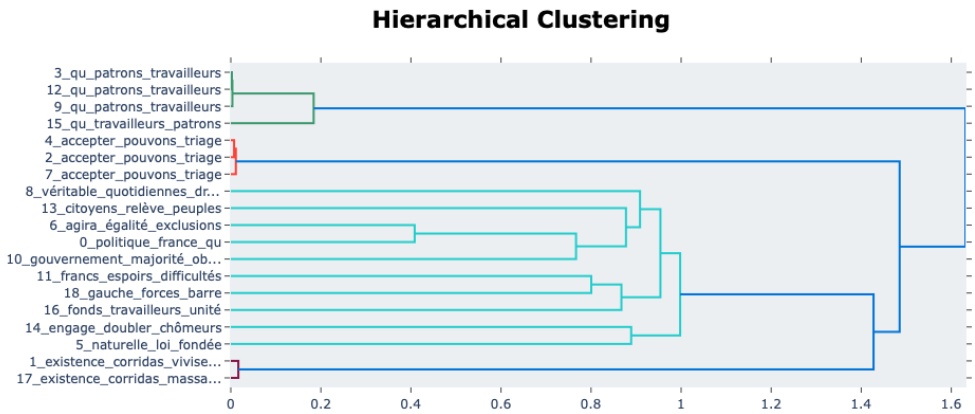


Figure 3: Hierarchical cluster

the corpus document. The high concentration in this rather generic cluster (whose keywords are "politique", "france", "droite", "pays", "gauche") reveals an important limitation. This means that the electoral manifestos are very homogeneous in their political register and that UMAP dimension reduction and HDBSCAN clustering can't differentiate them into more precise thematic groups. **Topics 3, 9, 12, 14, 15 mention employment and are thus used for the rest of the analysis.**

Table 2: Topics from BERT Model (Top 8 keywords)

Topic	Count	Representation
-1	47	voulons, qu, kofi, rôle, système, paris, vie, société
0	5052	politique, france, qu, droite, ans, pays, faire, français
1	116	existence, corridas, vivisection, véto, minimun, massacres, traitements, disposant
2	44	accepter, pouvons, triage, illusion, non, accepte, sortir, temps
3	38	qu, patrons, travailleurs, maintenir, payer, patronat, faire, devons
4	27	accepter, pouvons, triage, illusion, non, accepte, sortir, temps
5	27	naturelle, loi, fondée, cohérence, connaissance, éliminer, programme, védique
6	23	agira, égalité, exclusions, développement, modernisera, veillera, travaillera, garantira
7	23	accepter, pouvons, triage, illusion, non, accepte, sortir, temps
8	22	véritable, quotidiennes, droite, indispensable, conviction, cni, difficultés, minée
9	22	qu, patrons, travailleurs, maintenir, payer, patronat, faire, devons
10	21	gouvernement, majorité, objectifs, campagne, présidentielle, mai, ouverture, électorale
11	20	francs, espoirs, difficultés, traditions, cœur, parti, candidats, communiste
12	19	qu, patrons, travailleurs, maintenir, payer, patronat, faire, devons
13	18	citoyens, relève, peuples, mouvement, passe, fruits, européenne, vis
14	15	engage, doubler, chômeurs, compétitivité, production, 000, personnes, paris
15	15	qu, travailleurs, patrons, maintenir, payer, emplois, patronat, faire
16	12	fonds, travailleurs, unité, solution, école, sou, friche, ouvrière
17	12	existence, corridas, massacres, vivisection, respectées, minimun, traitements, généralisé
18	11	gauche, forces, barre, ministres, affirmé, droite, rassemblement, alliance

Together, **these five topics contain 92 documents, or 4.3% of the corpus**, so a proportion quite close to that obtained with LDA. The hierarchical clustering visualization 3 reveals an interesting structure: topics 3, 9, 12, and 15 are tightly grouped (near-zero distance), suggesting that they might be based on the same single textual template (the Lutte Ouvrière manifesto). This raises a methodological concern because these topics may reflect the repetition of a single party tract across multiple constituencies rather than a true thematic diversity in how unemployment is discussed.

4.2 Correlation analysis

Using LDA then BERTopic, we then study the correlation between the effective unemployment rate and the intensity with which political manifestos mention thematics related to employment. To do so we define topic intensity as the probability assigned to Topic 5 by LDA and as the sum of probabilities assigned to clusters 3, 9, 12, 14, and 15 by BERTopic. The results are presented in Table 3

Table 3: Correlation between political orientation and topic distributions

Political orientation	n depts	LDA r	LDA p-value	BERTopic r	BERTopic p-value
Left (PS, Greens)	77	-0.001	0.992	-0.038	0.740
Far-left (PCF, LO)	73	0.127	0.285	0.217	0.065
Right (RPR, UDF)	89	0.078	0.468	-0.039	0.720
Far-right (FN)	73	0.086	0.470	0.147	0.214

The results show a consistent pattern: correlations are weak to negligible across all political orientations, and none reach usual significance thresholds ($p < 0.05$). The strongest signal is observed for the Far-left (primarily PCF and Lutte Ouvrière) with BERTopic with signifiance just above the 5% threshold but below the 10% one. The scatter plots (Figure 4 and Figure 5) visually confirms this: the far-left regression slope is the steepest, while all other political parties show no correlation. For the Far right (Front National), a modest positive correlation appears with BERTopic ($r = 0.147$) but is far from significant ($p = 0.214$). A complementary view of these results is provided by the mean topic intensity scores by orientation (Table 4). Far left candidates display by far the highest intensity with both LDA and BERTopic (LDA: 0.234; BERTopic: 0.297), approximately 10 to 15 times higher than right and left candidates. This confirms that far-left tracts are not only more correlated with local unemployment but discuss unemployment with much greater frequency and intensity overall.

Table 4: Mean topic intensity by political orientation for LDA and BERTopic models.

Political orientation	Mean LDA intensity	Mean BERTopic intensity
Far left	0.234	0.297
Left	0.014	0.116
Other	0.026	0.115
Right	0.016	0.090
Far right	0.008	0.096

We can say that LDA and BERTopic identify some consistent results about far-left association between discourse and unemployment realities. However, the models disagree on the other political parties, and results aren't significant, which limits generalization.

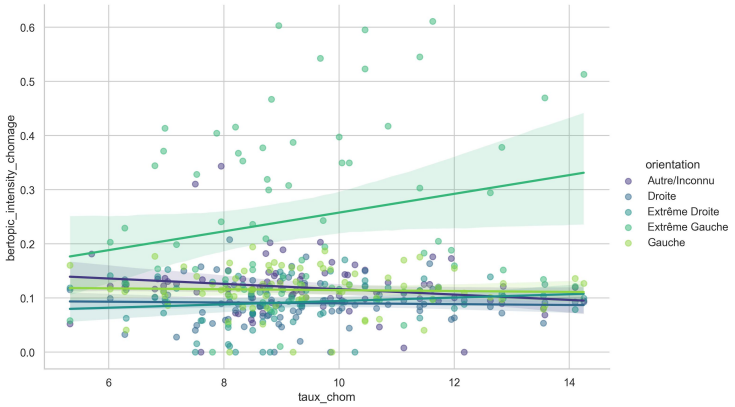


Figure 4: Correlation between effective unemployment rate and intensity of discourse on employment (BERTopic

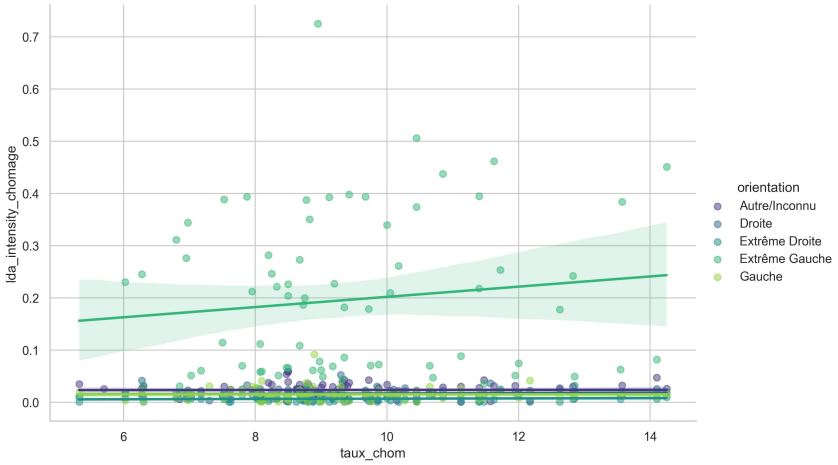


Figure 5: Correlation between effective unemployment rate and intensity of discourse on employment (LDA)

5 Conclusion and limitations

This study intends to give a quantitative test of the economic voting hypothesis applied to the supply side of politics: do candidates adapt their discourse to local economic conditions? Our findings suggest they largely do not. Despite a significant rise in unemployment in the 1970s, 1980s and 1990s, the intensity of unemployment-related discourse in electoral manifestos shows little to no correlation with departmental unemployment rates across the political spectrum. This is consistent with theories emphasizing the primacy of national party cues over local economic signals in legislative campaigns [3]. The exception of the far left is explained by the ideological homogeneity of Lutte Ouvrière's campaigns than by genuine economic responsiveness. Future work could disentangle national messaging effects from local adaptation by comparing within-party variation across departments with contrasting economic profiles.

Having said all that, we must face the numerous limitations of our work. First, and maybe most importantly, none of the correlations we analyze are statistically significant at the 5% level, whether it be for LDA or BERTopic. The statistical power is also limited by relatively small samples at the department level. Second, the result for far left are essentially those driven by Lutte Ouvrière tracts. This suggests that perhaps what is detected locally is not relative to local unemployment rates but to local implementation of the party. Third, the 32% of manifestos classified as 'Autre/Inconnu' due to incomplete party label matching in the metadata represents a big loss of information. If these unclassified tracts are significantly different in their unemployment discourse from classified ones, the orientation-level results could be biased. Finally, the near-total dominance of the cluster 0 of BERTopic (84.7% of documents) signals that the corpus is too lexically homogeneous for BERTopic to produce a well-differentiated topic structure. We could argue that this is, in and of itself, a scientific finding. In fact it corroborates the 'langue de bois' theory suggesting that political discourse is relatively generic and doesn't adapt to local and temporally situated issues. Yet, this definitely limits correlation findings.

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